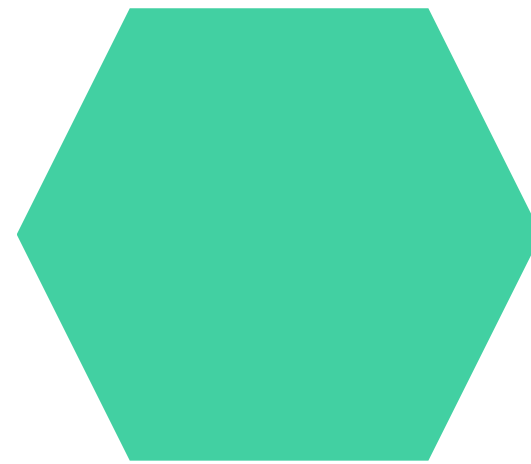


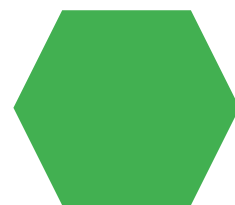


Project Title - Seasonal Agriculture Performance Analysis

Student Name- SAMRUDDHI PRADEEP REPAL

College Name- JAYAWANTRAO SAWANT POLYTECHNIC

AICTE STU ID : STU6a203feff1e631780498415



PROBLEM STATEMENT

Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description

This project focuses on analyzing agricultural performance across different seasons using data analytics and visualization techniques.

The dataset contains 4,000 agricultural records and 28 variables, covering different:

- Seasons
- Crops
- States and districts
- Farming practices
- Environmental conditions
- Irrigation methods
- Resource usage
- Yield
- Profitability
- Disease and pest risk

Analysis Process

1. Load and understand the dataset
2. Check data structure and data types
3. Identify missing values and duplicates
4. Clean and prepare the data
5. Compare agricultural performance across seasons
6. Analyze crop and irrigation patterns
7. Study relationships between variables
8. Visualize important findings
9. Draw conclusions
10. Provide data-driven recommendations



WHO ARE THE END USERS?

The insights generated from this analysis can be useful for:

- Farmers - to understand seasonal performance and improve planning
- Agricultural Planners - to plan crops and resources according to seasonal conditions
- Agricultural Officers - to identify areas requiring attention and support
- Researchers & Students - to study agricultural trends and relationships
- Agri-businesses - to understand crop performance and profitability
- Policy Makers - to support evidence-based agricultural planning

Technology Used

1. Python

Used as the primary programming language for data analysis.

2. Pandas

Used for data loading, cleaning, grouping and statistical analysis.

3. NumPy

Used for numerical calculations and data processing.

4. Matplotlib

Used to create charts and visualize agricultural trends.

5. Jupyter Notebook

Used to document and execute the complete data analysis workflow.

6. Microsoft PowerPoint

Used to present the project findings and insights.

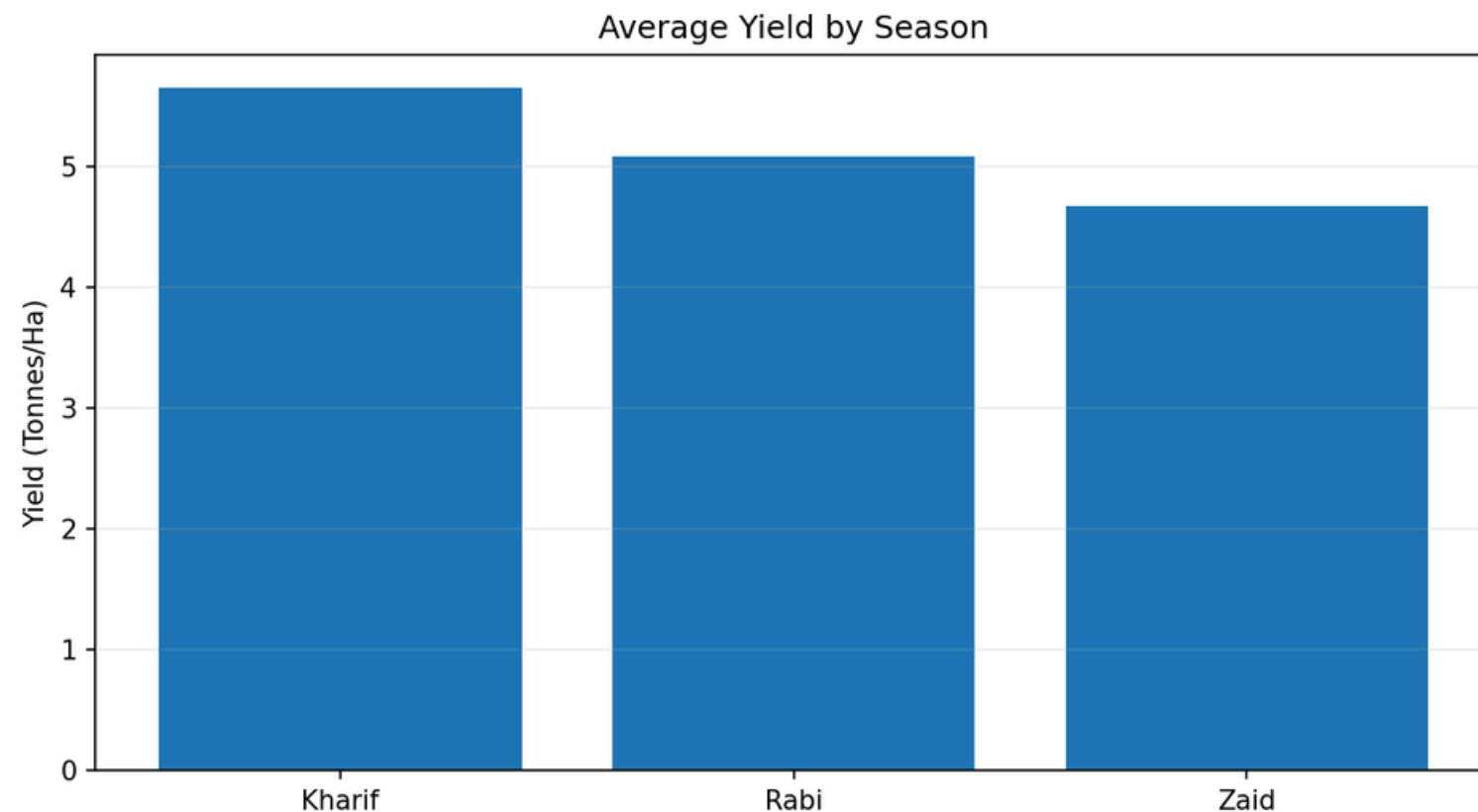
7. Dataset

- Records: 4,000
- Variables: 28
- Seasons: Kharif, Rabi, Zaid
- Duplicate records: 0
- Missing cells: 120



RESULTS

Seasonal Yield Comparison



Kharif has the highest average yield at 5.64 t/ha, followed by Rabi (5.08) and Zaid (4.67).

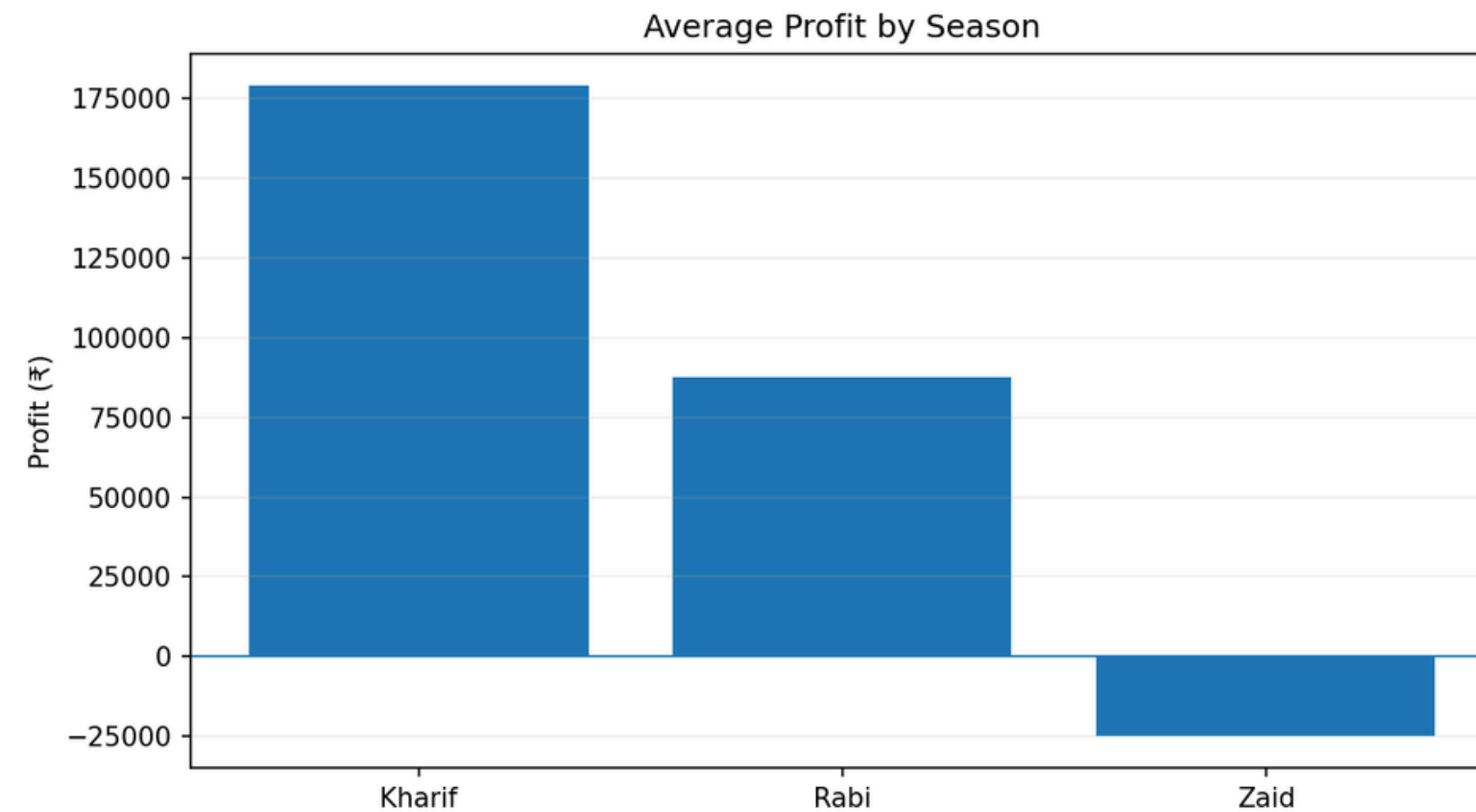
Analysis code used:

```
# Load and inspect the dataset
import pandas as pd

df = pd.read_csv("seasonal_agriculture_performance_dataset.csv")
print(df.shape)
print(df.info())
print(df.isnull().sum())
```

RESULTS

Seasonal Profit Comparison



Average profit is highest in Kharif (₹178,915), falls in Rabi (₹87,689), and is negative in Zaid (-₹24,805).

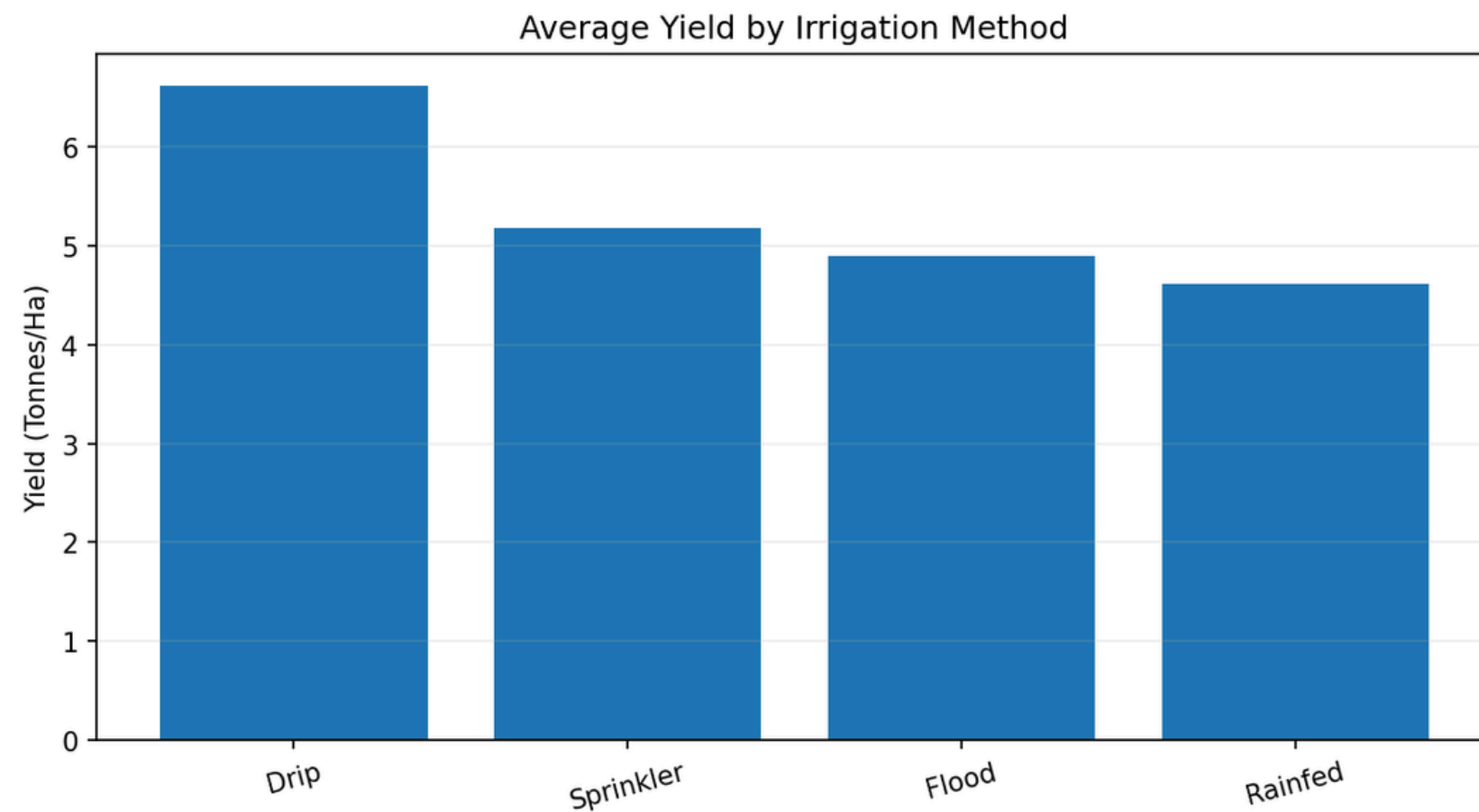
Analysis code used:

```
# Seasonal performance comparison
season_summary = df.groupby("Season").agg(
    avg_yield=("Yield_Tonnes_Ha", "mean"),
    avg_profit=("Profit_INR", "mean"),
    avg_water_eff=("Water_Efficiency_t_per_1000m3", "mean")
).round(2)

print(season_summary)
```

RESULTS

Irrigation and Yield



Drip irrigation records the highest average yield (6.62 t/ha) and highest average profit (₹219,626) among the irrigation methods in the dataset.

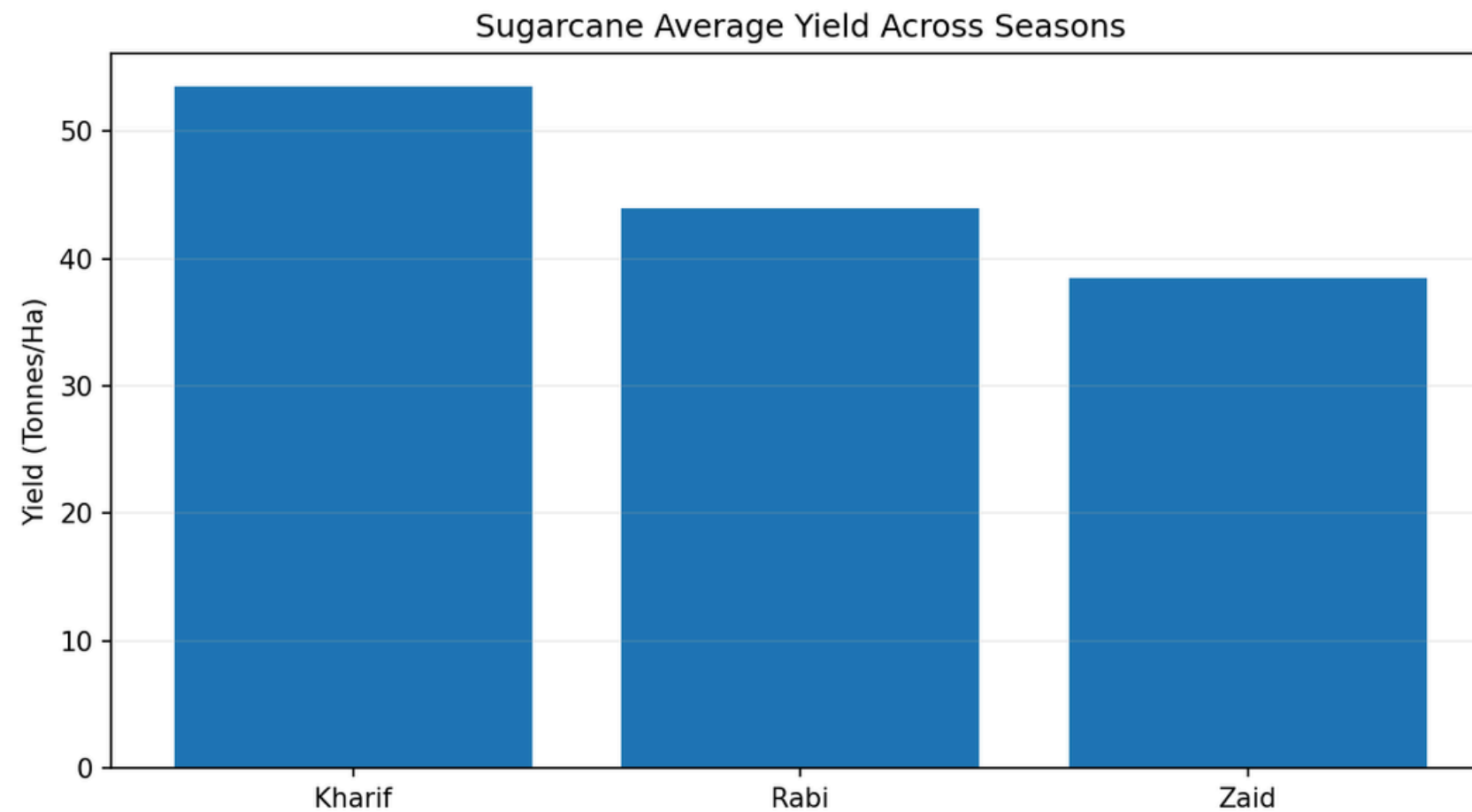
Analysis code used:

```
# Relationship analysis
numeric = df.select_dtypes("number")
corr = numeric.corr()

print(corr["Yield_Tonnes_Ha"]
      .sort_values(ascending=False)
      .head(8))
```

RESULTS

Crop-Level Seasonal Pattern



Sugarcane has substantially higher average yield than the other crops in every season; its Kharif average is about 53.46 t/ha.

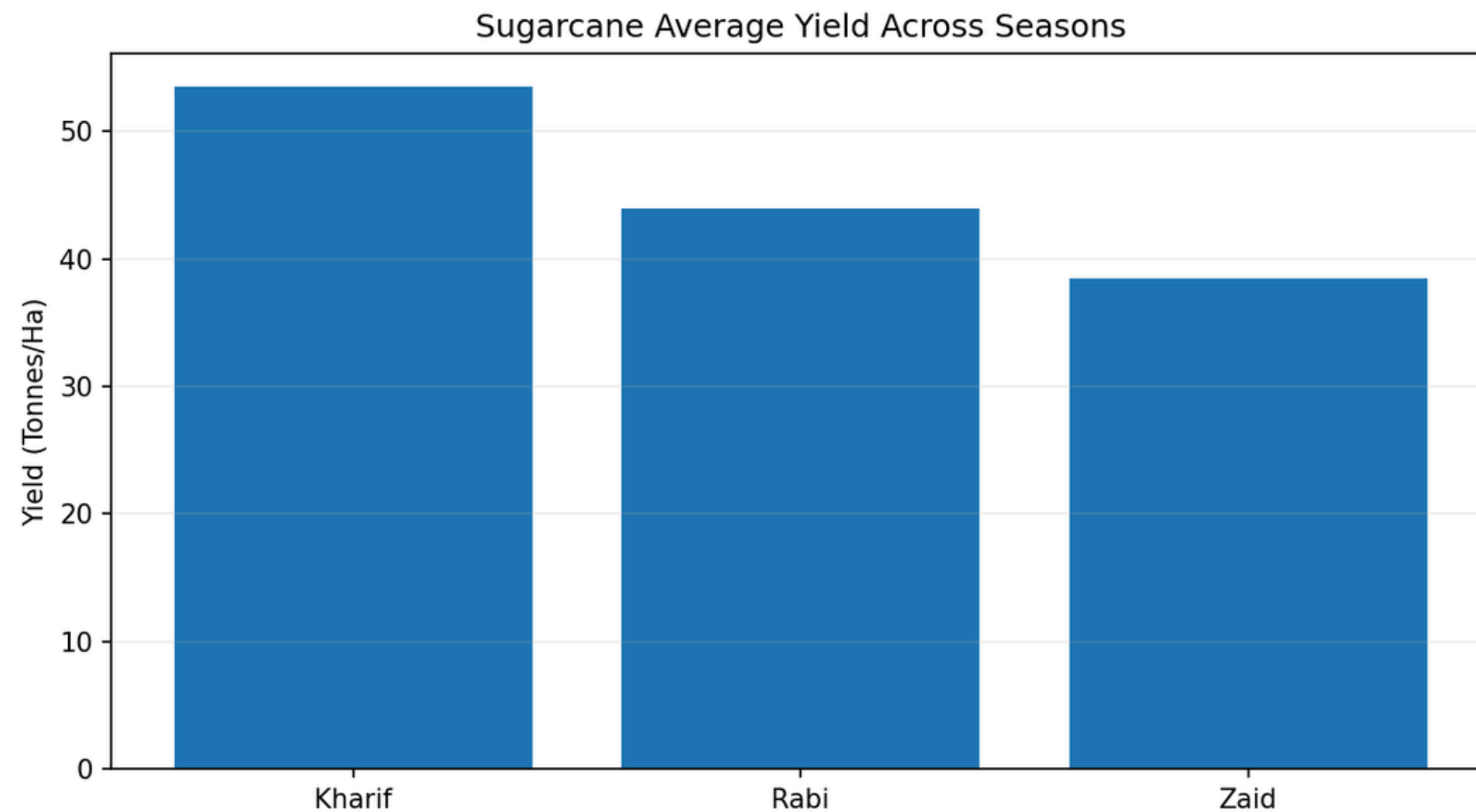
Analysis code used:

```
# Irrigation comparison
irrigation_summary = df.groupby("Irrigation_Method").agg(
    avg_yield=("Yield_Tonnes_Ha", "mean"),
    avg_profit=("Profit_INR", "mean"),
    water_eff=("Water_Efficiency_t_per_1000m3", "mean")
).round(2)

print(irrigation_summary)
```

RESULTS

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).round(2)

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```

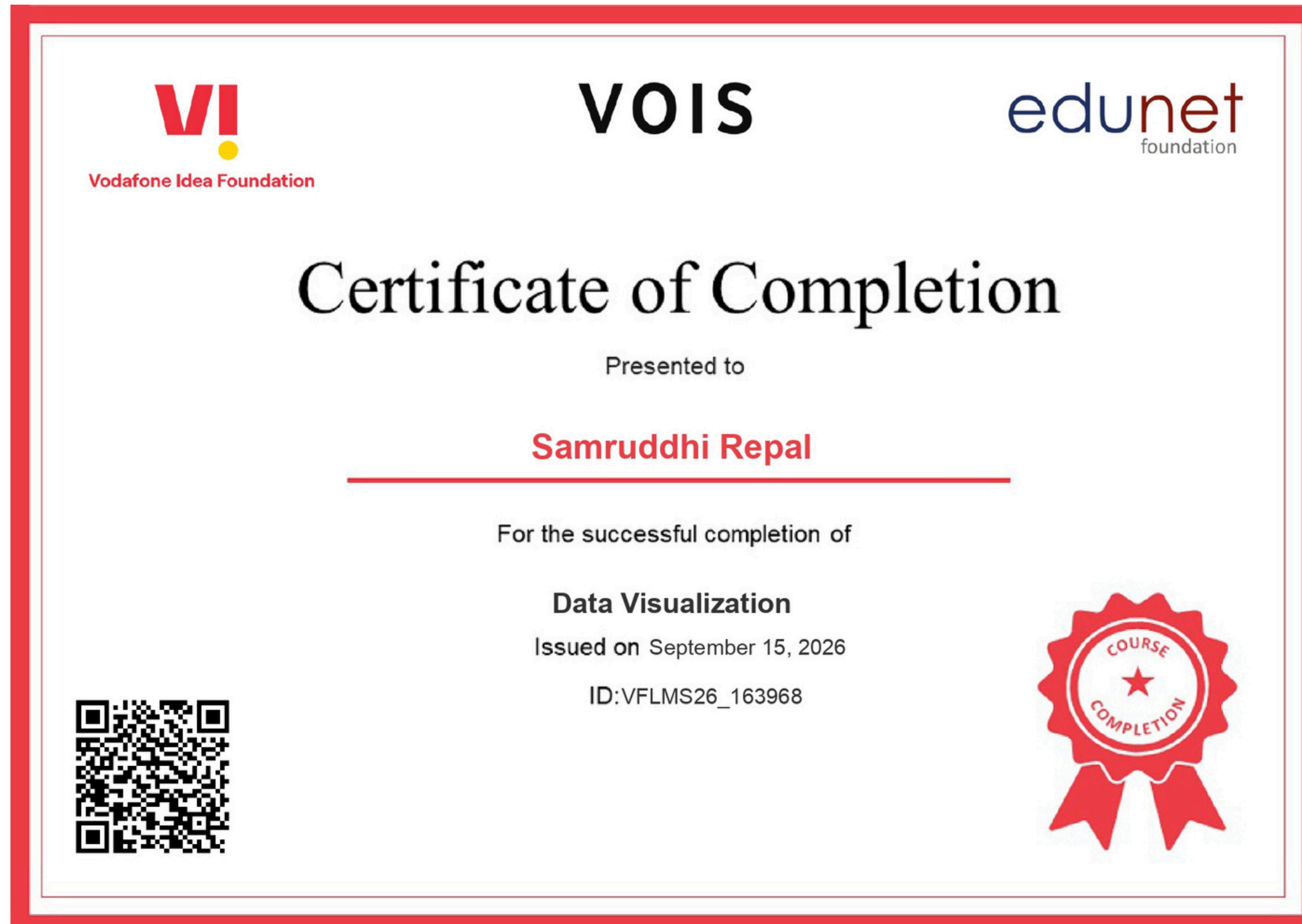
Future scope

The project can be further enhanced by:

- Developing machine learning models to predict crop yield.
- Forecasting agricultural profitability for future seasons.
- Integrating real-time weather and rainfall data.
- Adding soil-quality and fertilizer information.
- Incorporating real-time agricultural market prices.
- Developing an interactive dashboard for farmers and agricultural planners.
- Using geographical maps for district/state-level analysis.
- Creating early-warning systems for disease and pest risks.
- Analyzing year-to-year seasonal trends using historical datasets.
- Developing crop recommendation systems based on seasonal conditions.



VOIS Course completion certificate - Course name- Data Visualization



Thank you

Seasonal Agriculture Performance Analysis

Presented by:
SAMRUDDHI PRADEEP REPAL